LEARNER GUIDE

Both sides, and brackets

Equations, Inequalities and Simultaneous Relationships · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Until now the unknown x sat on one side of the equals sign. Now it can appear on **both** sides, or be locked inside a **bracket**. The trick is to **tidy first**. If x is on both sides, **collect** it onto one side by subtracting the smaller number of x's from both sides. If there is a bracket like 3(x + 2), **expand** it first — the outside number multiplies everything inside, giving 3x + 6. Once tidied, it is an ordinary balance problem you already know how to solve, and you still **check** by substituting back.

What you are learning to notice

- **Like terms** only add to like terms: x's with x's, numbers with numbers.
- Subtracting the same number of x's from **both sides** collects the unknown — the same 'do it to both sides' rule.
- A bracket a(x + b) means a rectangle: it expands to **ax + ab** (multiply everything inside).
- After tidying, it is a two-step equation — and the substitution **check** still proves the answer.

Moving through it, one step at a time

First tidy. If there is a bracket, expand it: multiply the outside number by each term inside. If x appears on both sides, subtract the smaller x-term from both sides so the unknown is on one side only. Now you have a familiar equation: clear the number by adding or subtracting on both sides, then divide. Finally, substitute your answer into the original equation and check both sides are equal. Show every line.

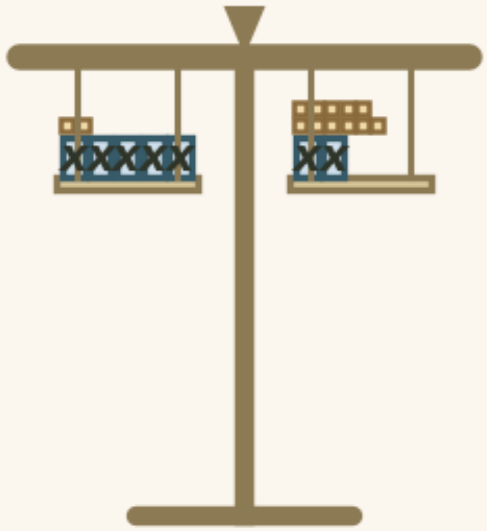
Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — The unknown on both sides

Solve $5x + 2 = 2x + 11$ — there are x's on both sides.

$$5x + 2 = 2x + 11$$

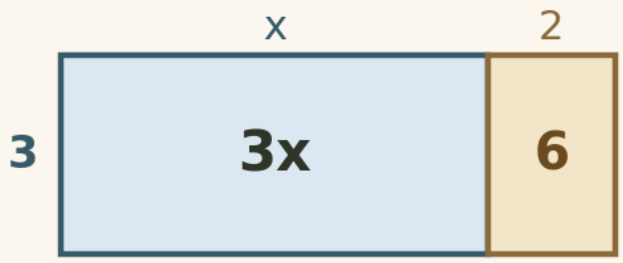


1. Collect x: subtract $2x$ from both sides.	$5x + 2 - 2x = 2x + 11$ $- 2x$
2. The x is now on one side only.	$3x + 2 = 11$
3. Subtract 2 from both sides.	$3x = 9$
4. Divide both sides by 3.	$x = 3$
5. Check in the original.	$5\times3 + 2 = 2\times3 + 11$ $= 17$ ✓

$x = 3$ — subtract $2x$ to collect the unknown, then solve $3x + 2 = 11$; the check gives 17 on both sides.

Example 2 — A bracket to expand

Solve $3(x + 2) = 18$ — the x is inside a bracket.



1. Expand: the 3 multiplies both x and 2.	$3(x + 2) = 3x + 6$
2. Rewrite the equation.	$3x + 6 = 18$
3. Subtract 6 from both sides.	$3x = 12$
4. Divide both sides by 3.	$x = 4$
5. Check in the original.	$3(4 + 2) = 18$ ✓

$x = 4$ — expand to $3x + 6 = 18$, then solve; the check $3(4 + 2) = 18$ confirms it.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (collecting like terms, expanding brackets, and keeping an equation balanced (EQ-01)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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